MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6286

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, high frequency oscillator

Outline --- 2-1 (EIA)

Base --- Pinch press (4-leads in line)

Envelope --- T2X3

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4
Element	---	a (red dot)	-f	g	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ik	Pp	Ⓒ Alt
Unit:	Vdc	Vdc	Vdc	mAdc	W	ft
Maximum:	1.50	100	---	7.0	0.45	Note 1
Minimum:	1.00	---	---	---	---	---
TEST CONDITIONS:	1.25	67.5	-2.0	---	---	---

GENERAL:

Qualification - Required

Ⓒ denotes changes

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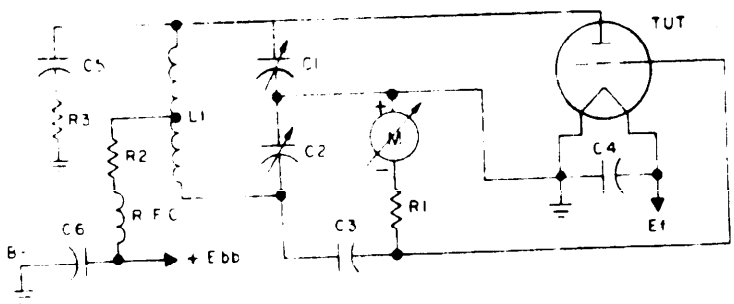
METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
(C) 1202	Filament burnout	-		---	---	---	---	---	---
	Quality conformance inspection, part 1								
(C) 1266	Total grid current	2		0.65	II	Ic	0	-0.5	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	4	8	mAdc
1306	Transconductance (1)	-		0.65	II	Sm	1,700	2,500	µmhos
1237	Oscillator grid current (3)	2, 3	Ef = 1.0 Vdc; Ebb = 50 Vdc; F = 25 MHz	0.65	II	Ic	100	---	µAdc
(C) 1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Rp = 2,000 ohms	---	---	Ep	---	25	mVac
1301	Filament current	-		---	---	If	115	135	mAdc
1306	Transconductance (2)	-	Ef = 1.0 Vdc	---	---	Sm	1,400	---	µmhos
1316	Amplification factor	-		---	---	Mu	10	13	---
1237	Oscillator grid current (1)	3	Ebb = 67.5 Vdc; F = 25 MHz	---	---	Ic	220	---	µAdc
1237	Oscillator grid current (2)	3	Ef = 1.0 Vdc; Ebb = 67.5 Vdc; F = 25 MHz	---	---	Ic	100	---	µAdc
1331	Direct-interelectrode capacitance	-	No shield	---	---	Cgp Cin Cout	1.2 1.0 1.6	2.0 1.6 2.6	pF pF pF
1116	Lead fatigue	-		4.0	S3	---	3	---	Arcs
(C) 1211	Insulation of electrodes	-		4.0	S3	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
(C) 2126	Glass strain	-		4.0	I	---	---	---	---
(C) 1119	Glass-to-lead seal strain	-		---	---	---	---	---	---
(C) 1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	4	Group A	---	---	---	---	---	---
---	Life-test end points (500 hours):								
1306	Transconductance (1)	-		---	---	Sm	1,300	---	μmhos
1237	Oscillator grid current (1)	-		---	---	Ic	110	---	μAdc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Test per circuit below:



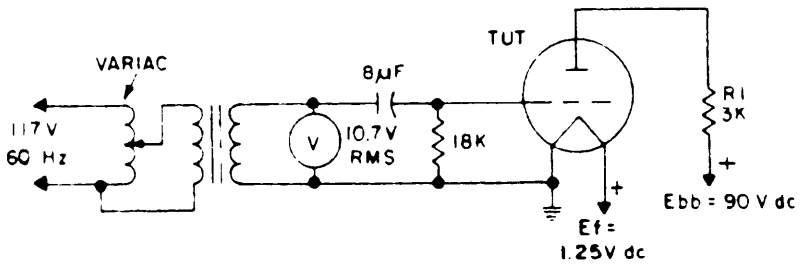
Circuit component values

C1 Variable capacitor adjusted to 45 pF ± 2-1/2%
C2 Variable capacitor adjusted to 65 pF ± 2-1/2%
C3 50 pF ceramic capacitor
C4 250 pF ceramic capacitor
C5 50 pF ceramic capacitor
C6 1,000 pF ceramic capacitor
R1 18,000 ohms ± 5% - 1/2 watt carbon resistor
R2 1,500 ohms ± 5% - 1/2 watt carbon resistor
R3 3,500 ohms ± 1% - 1/2 watt carbon resistor
RFC Ohmite Z-50 R.F. Choke
L1 12 turns of No. 18 enameled copper wire wound on 1/2 inch diameter lucite rod tapped at 7 turns from anode end with spacing between turns adjusted to produce oscillator frequency of 25 MHz.

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NOTES: -Continued

4. Dynamic life-test circuit:



Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2563)

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/270C ELECTRON TUBE, RECEIVING TYPE 6286

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-PM 1068-68

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